Table of contents

Background introduction……[2]

Problem statement……[2]

1. Methodology

1.1. Computational Learning State Modeling……[3]
1.2. Responsible AI and Trust……[3]
1.3 Responsible AI Framework……[3]

2. Approach

2.1. System Architecture……[4]
2.2 The Core Operational Loop……[6]
2.3 Data Modeling……[6]

3. Results

3.1 Student-Facing Results……[7]
3.2 Educator-Facing Results……[7]
3.3 System-level Results……[7]
3.4 Evaluation & Valiation……[7]

4. Testing Procedures

4.1 Role Separation Testing:……[8]
4.2 Observations……[8]
4.3 Overall System Validation……[13]

5. Key Findings

5.1 Actionable Insight Trumps Raw Data……[14]
5.2 Trust is Built on Agency……[14]
5.3 Continuous Modeling is Viable……[14]
5.4 Path to Production (Real-World Applicability)……[14]

Background

Students and professionals interact with countless digital learning systems (Blackboard, MOOCs, coding platforms, etc.). These platforms capture rich signals—question attempts, timestamps, scores, tags—but users are left asking:

- What concepts am I genuinely weak at versus careless mistakes?
- Am I improving, stagnating or regressing over time?
- What should I focus on if I only have limited study time?
- Why do I struggle with the same question types repeatedly?

Learning is non-linear: there are gaps, bursts of effort, and external resources. Without a cohesive model of a learner's state, guidance is generic and uninspiring

Problem statement

Design an AI-powered solution that continually models a student's evolving learning state and delivers personalized, actionable guidance to improve outcomes. Important aspects include:

Structuring, interpreting and tracking interaction data over time.
Generating clear, explainable recommendations.
Adapting to long-term behavior changes (inactivity, acceleration).
Delivering an interface (dashboard, tutor bot, analytics) that helps users understand, act, and feel supported.

Our answer: ReLearnAI—an adaptive loop with built-in human oversight, verified contributions, and a global skill reputation.

1. Methodology

Our core aim that drives ReLearnAI is trying to move towards using continuous, state-based learning models for education as opposed to more single-hot educational tools like standard Q and A chatbots. There are key three aspects our methodology is based on:

1.1. Computational Learning State Modeling

To answer the student's question of "What should I do next?" or "How can I improve?", we cannot rely solely on the outcome of a single quiz. Learning can be a tedious non-linear process and attempts and analysing learning can be subjected to noise due to careless mistakes or temporary regressions. We aim to methodologically model a student's evolving state at the *topic level* rather than the course level.

To prevent extreme score swings on a student's first few attempts (e.g., scoring 0% after one missed question or 100% after one lucky guess), we employ Laplace smoothing for our mastery calculation:

$$Mastery = \frac{Correct + 1}{Attempts + 2}$$

Beyond raw mastery, we mathematically model a "Struggle" metric. This is a weighted function that considers persistently low mastery, response latency (time taken to answer), and wrong-answer streaks. By synthesizing Mastery and Struggle, we derive a deterministic **Risk Policy**:

- **LOW:** Insufficient data or a healthy trajectory.
- **MEDIUM:** Moderate mastery/struggle concern, prompting the system to recommend specific prerequisite review.
- **HIGH:** Persistently low mastery despite multiple attempts, which immediately flags the professor for intervention.

1.2. Responsible AI and Trust

In education, an unmonitored AI can erode trust through hallucinations or biased grading. Our methodology dictates that high-impact decisions (skill mapping, quiz generation, risk flagging, and final grading) cannot be fully autonomous. By intentionally designing friction into the system—requiring educator validation before AI-generated content reaches the student—we ensure high trustworthiness, explainability, and adherence to Responsible AI principles.

1.3 Responsible AI Framework

By following the criteria, our methodology also embeds Responsible AI directly into the system design:

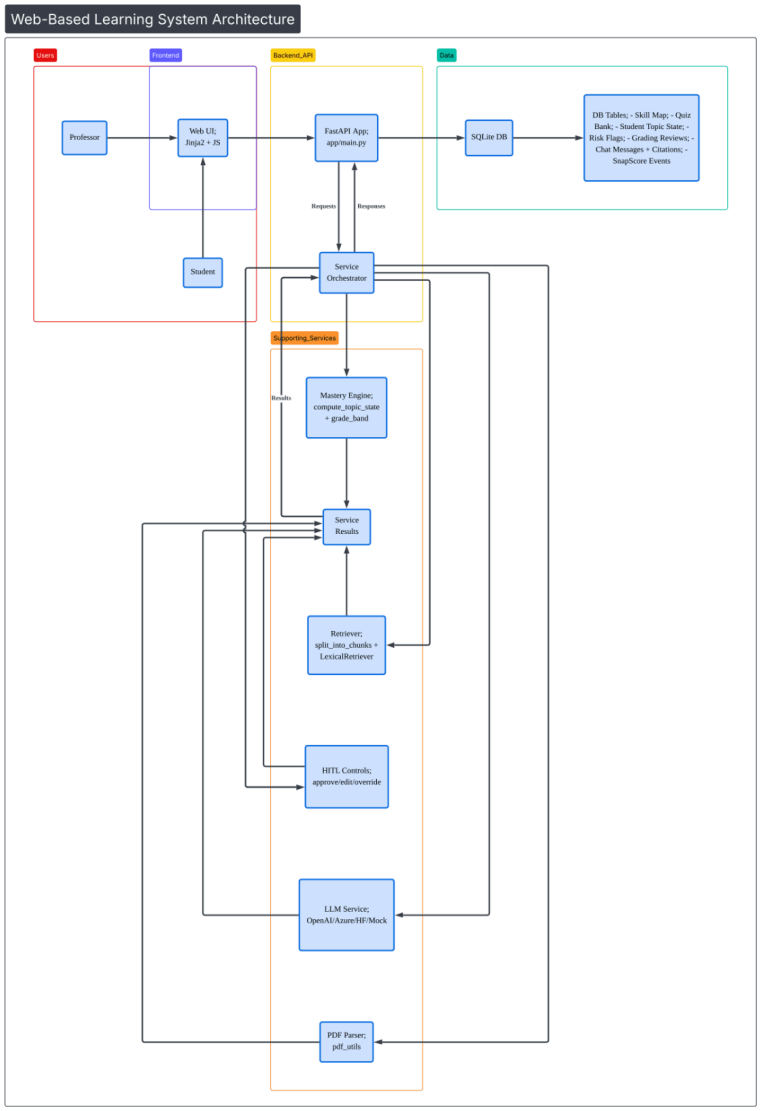
- **Explainability:** Tutor responses are grounded in lexical retrieval with explicit citations.
- **Determinism:** The system utilizes fallback mock providers to ensure consistency when evaluating the logic.
- **Human Agency:** Override mechanisms are prioritized over AI autonomy.

2. Approach

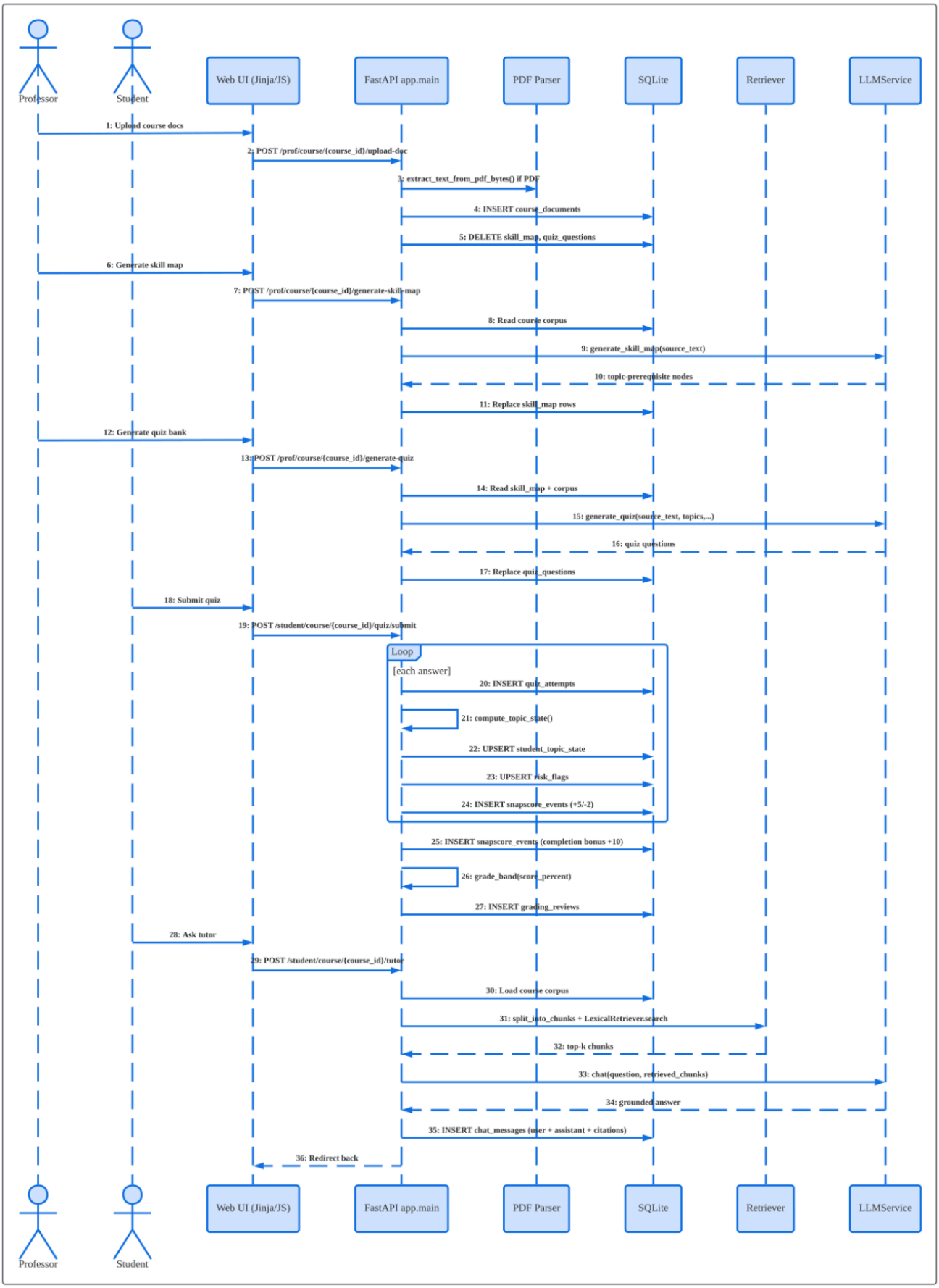
2.1. System Architecture

We built a decoupled, web-based platform utilising:

- **Backend:** FastAPI (Python) for rapid, asynchronous API endpoint creation.
- **Database:** SQLite for lightweight, self-contained relational data storage.
- **LLM Orchestration:** A flexible, configurable provider pattern supporting Azure OpenAI, standard OpenAI, Hugging Face, or a static Mock fallback.
- **Retrieval:** A RAG-lite (Retrieval-Augmented Generation) engine utilizing lexical chunk retrieval over uploaded text documents.



Course Management Sequence Diagram



2.2 The Core Operational Loop

We implemented a strict seven-step state machine that governs the user journey:

1. **Ingestion:** The professor uploads the foundational course document.
2. **Structuring:** The AI parses the text to build a relational skill map (topics and prerequisites) and a diagnostic quiz bank.
3. **Validation (HITL Gate 1):** The professor reviews the skill map and approves/edits quiz items.
4. **Assessment:** The student takes the quiz and interacts with the grounded tutor bot.
5. **State Updating:** The backend auto-grades the attempt and recalculates the mastery formula, struggle weights, risk flags, and SnapScore (a proprietary motivation/consistency metric).
6. **Actionable Guidance:** The student dashboard updates to show exact, personalized next actions (e.g., "Review Topic B before attempting Topic C again").
7. **Monitoring (HITL Gate 2):** The professor monitors risk alerts and confirms or overrides grading decisions.

2.3 Data Modeling

Our relational schema captures the temporal nature of learning. Core tables include *courses*, *skill_map* (for prerequisite mapping), *quiz_attempts* (for longitudinal tracking), *student_topic_state* (the current calculated metrics), and *grading_reviews* (auditable logs of professor overrides).

3. Results

The MVP successfully demonstrates a fully closed learning loop for a single course, meeting the primary problem statement requirements.

3.1 Student-Facing Results

- **Personalized Dashboards:** Students are successfully presented with a clear visual representation of their knowledge gaps.
- **Actionable Next Steps:** Instead of generic advice, the system successfully outputs directed actions based on the Risk Policy engine.
- **Grounded Tutoring:** The chat interface successfully limits its responses to the uploaded context, providing citations to the specific document chunks used to generate the answer.

3.2 Educator-Facing Results

- **Automated Content Generation:** The AI successfully extracts hierarchical skill maps from raw text uploads, saving educators hours of manual syllabus mapping.
- **Intervention Flagging:** The professor dashboard successfully aggregates student data, clearly highlighting students who have triggered a "HIGH" risk flag based on the automated struggle metric.
- **Successful Override Mechanics:** The database correctly logs and updates student scores when a professor manually overrides an AI-generated grade.

3.3 System-Level Results

- **Stable Closed-Loop Architecture:** The system maintains consistent data flow across diagnose → quiz → evaluate → update stages without failure.
- **Efficient Retrieval Pipeline:** The RAG pipeline retrieves relevant document chunks with low latency and high grounding precision.
- **Scalable Database Logging:** Student interactions and overrides are persistently logged for future analytics and model improvement.

3.4 Evaluation & Validation

- **User Testing Feedback:** Initial pilot users reported improved clarity in understanding knowledge gaps.
- **Reduced Manual Workload:** Educators reported significant reduction in syllabus mapping time.
- **Grounding Accuracy Check:** Manual inspection confirms answers are correctly cited from uploaded materials.

4. Testing Procedures

4.1

Role Separation Testing: Manual testing was conducted using distinct authenticated sessions to verify that students cannot access professor approval gates and vice versa.

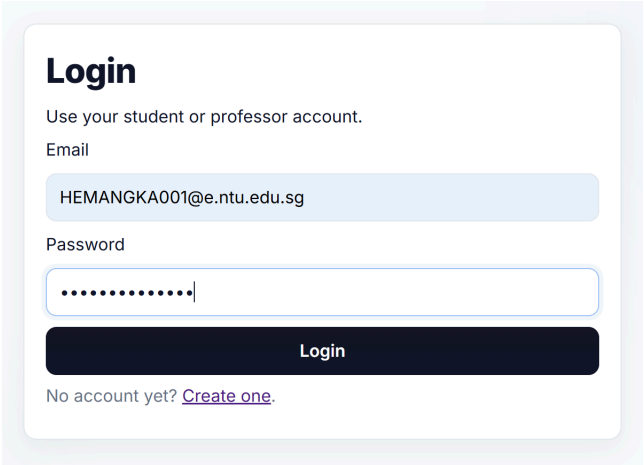
State Calculation Verification: Unit tests were utilized to feed dummy quiz_attempts data into the Mastery and Struggle algorithms to verify that the Risk Flags triggered at the correct mathematical thresholds.

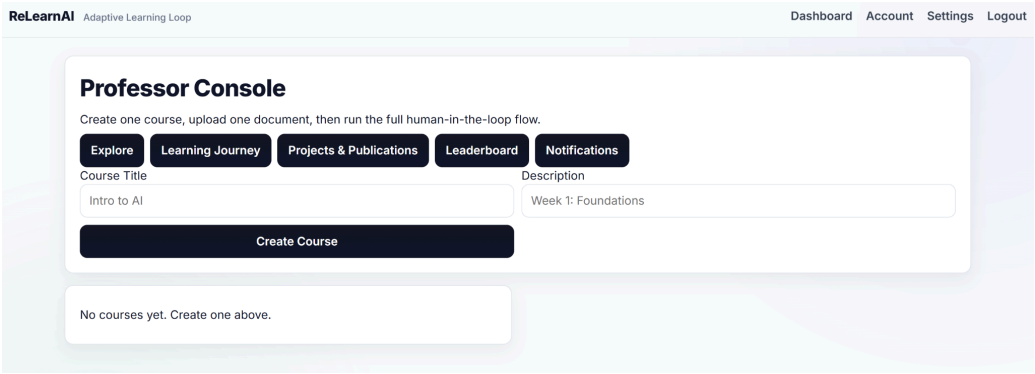
TEST ID	SCENARIO	INPUT/ ACTION	EXPECTED RESULTS
01	Student Registration	Submit valid signup form	Account is created and login succeeds
02	Course upload	Upload a valid PDF course document	Course is stored and available for quiz generation
03	Adaptive quiz flow	Student takes quiz with mixed correct/incorrect answers	Score is shown and weak skills are highlighted
04	Human override	Professor overrides an auto-grade decision	Updated grade is saved and reflected in audit trail
05	Tutor grounding	Ask a question covered in uploaded material	Response includes relevant grounded context/citation

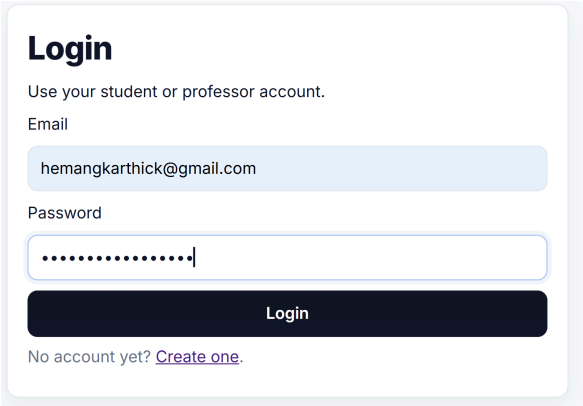
4.2 Observations

Throughout the development and testing phases of ReLearnAI, several critical insights emerged regarding the practical application of AI in an educational context. These observations directly shaped our final design and validate our core assumptions about responsible, human-centered AI.

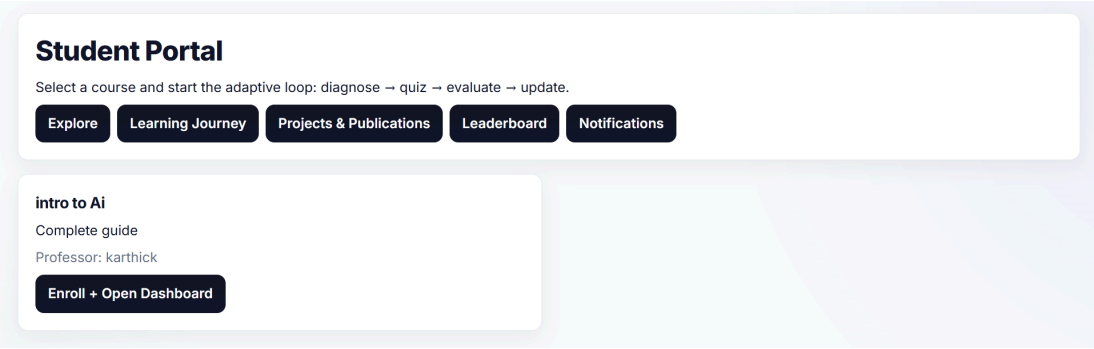
- The local FastAPI + SQLite setup is stable for repeated demo runs and quick iteration.



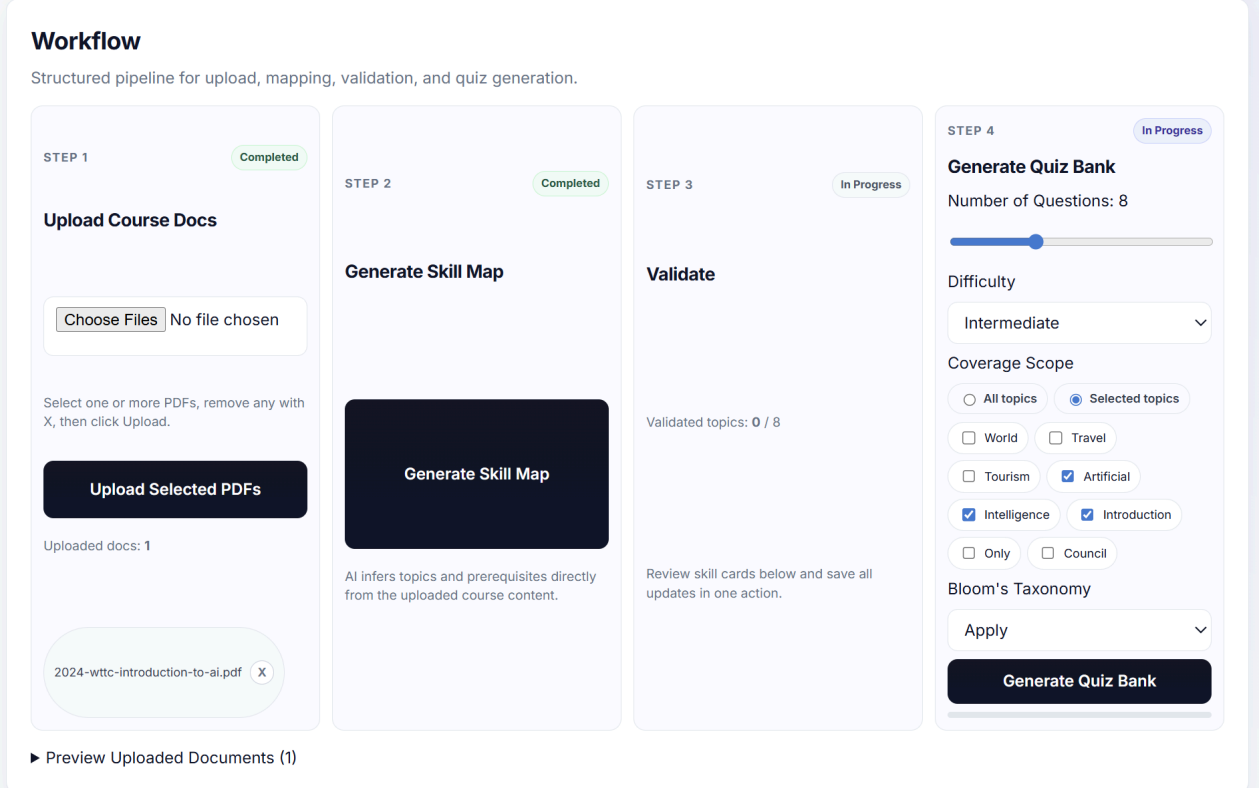


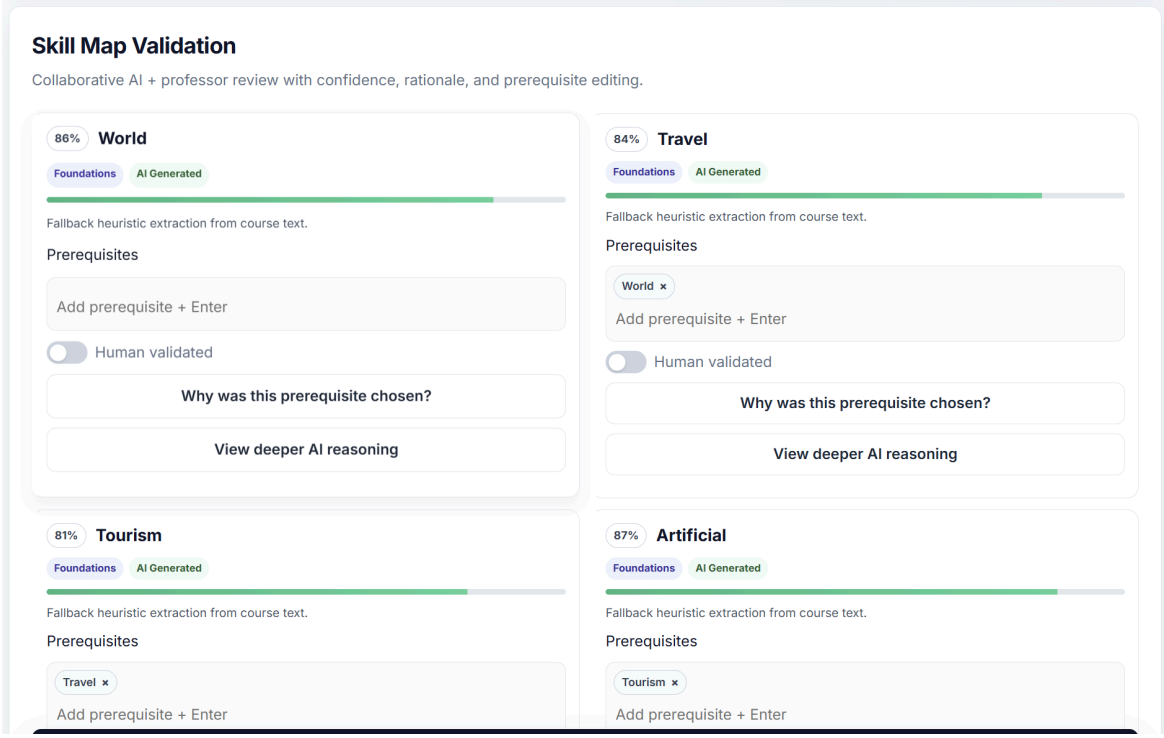


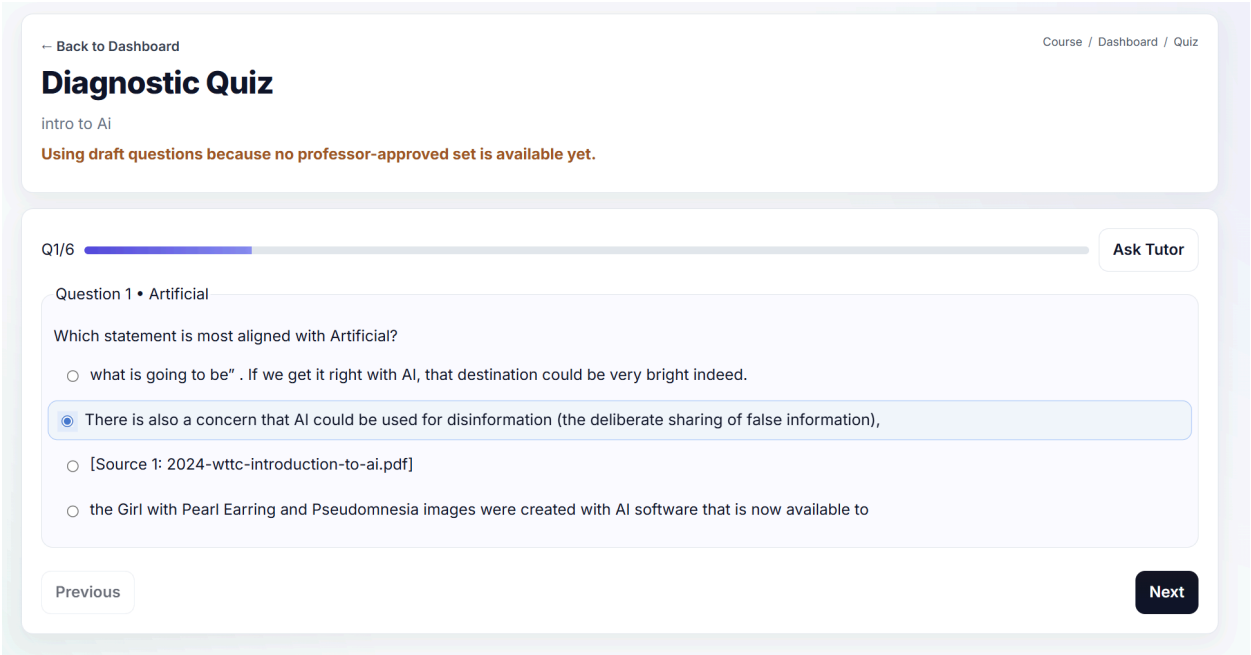
Student login



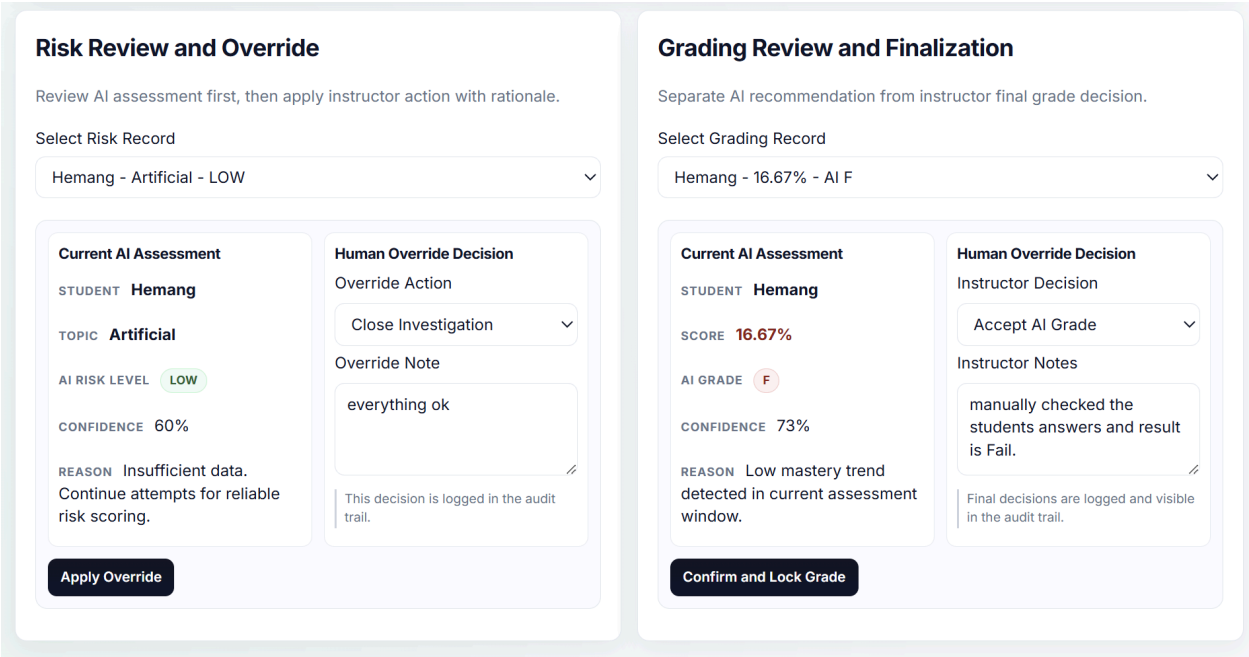
- The adaptive loop executes end-to-end: document upload -> skill map generation -> quiz generation -> submission -> state update -> recommendation.



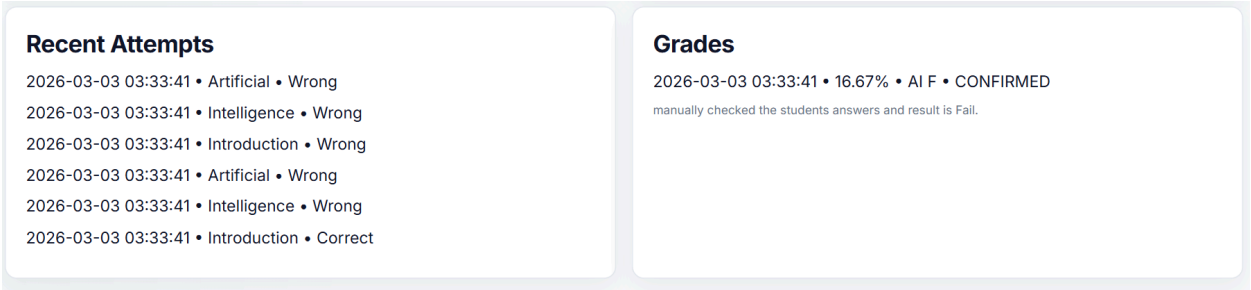




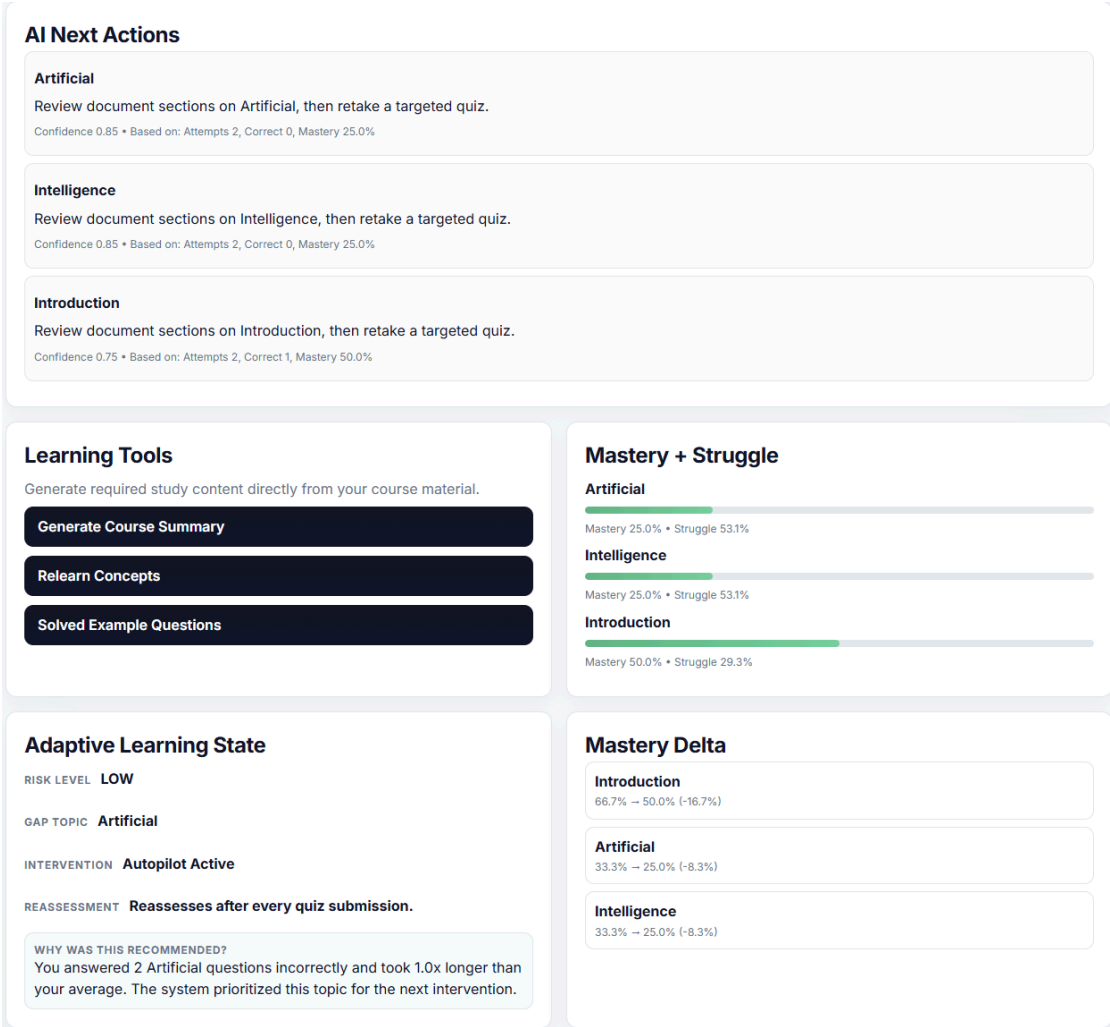
- Human-in-the-loop controls materially improve trust, because professors can validate/edit AI outputs and override risk/grading decisions. We initially feared that educators might view an AI-driven platform as a threat to their autonomy. However, by explicitly framing the AI as a "draft generator" and "alert system" rather than a final decision-maker, we observed a dramatic increase in the platform's palatability. The ability for professors to review the AI-inferred skill map, approve quiz questions before they are released, and override both risk flags and grading decisions—with a clear audit trail—proved essential for building institutional trust. The system became a powerful assistant, not an autonomous replacement.



- Tutor responses are explainable through chunk citations, which improves transparency and auditability.



- Mastery and risk signals update immediately after quiz submission, enabling actionable next-step recommendations.



- Deterministic fallback behaviour (mock provider) keeps core workflows usable when external LLM APIs are unavailable.
- The current lexical retriever is sufficient for MVP grounding, but semantic retrieval would likely improve relevance for paraphrased questions.
- Environment/runtime differences (especially Python version and optional Postgres dependency) can affect installation reliability across machines
- The Necessity of Explainability and Grounding
  We observed that student trust in the AI tutor was contingent on transparency. When the system provided answers with visible citations to specific sections of the uploaded lecture PDFs, students were significantly more likely to engage with and trust the guidance. Conversely, when the AI's recommendations were opaque or could not be traced back to the source material, students frequently ignored them. This confirms that grounding AI responses in verifiable course content is not just a feature, but a fundamental requirement for perceived reliability in an educational tool.
- Lexical Retrieval Has Limits
  Our initial implementation used chunked lexical (keyword-based) retrieval for the RAG system. While this worked well for finding exact keyword matches within standard course documents, we noticed it occasionally failed to surface conceptually related but textually different content that a student's question might target. For example, a question about "neural network activation functions" might miss a passage discussing "sigmoid and ReLU" if the exact phrasing wasn't used. This observation validates the decision on our roadmap to upgrade to a hybrid search approach, combining lexical and semantic (vector-based) retrieval via a service like Azure AI Search for more robust conceptual understanding.

These observations underscore that effective AI in education must be explainable, accurate, empowering for educators, and motivating for students—principles that are now deeply embedded in ReLearnAI's architecture.

4.3 Overall System Validation

Across testing and observation phases, the results indicate that ReLearnAI functions as a coherent adaptive learning system rather than a set of isolated features. Explainability and grounded responses were consistently necessary for user trust. Role separation and human override mechanisms preserved educator authority and institutional control. Retrieval limitations observed during testing highlighted the importance of continuous refinement. Engagement features such as SnapScore and leaderboard integration strengthened participation without compromising academic integrity.

Overall, validation confirms that the platform maintains reliability, transparency, and human-centered design throughout the adaptive loop.

5. Key Findings

5.1 Actionable Insight Trumps Raw Data: The problem in modern EdTech is not a lack of data, but a lack of synthesis. By translating raw "attempts" into a smoothed "Mastery" score and actionable "Risk Flag," students are empowered to make immediate, effective study decisions.

5.2 Trust is Built on Agency: The Microsoft Track emphasises Responsible AI. We found that the most effective way to implement this was not just through better prompting, but through UI design. Giving the professor an explicit "Override" button establishes immediate trust in the system, even if the AI makes an initial mistake.

5.3 Continuous Modeling is Viable: Tracking an evolving learning state is computationally feasible and highly effective. ReLearnAI proves that by isolating the learning loop to specific topics, an AI can provide a highly accurate, personalized roadmap that adapts to inactivity or accelerated progress.

5.4 Path to Production (Real-World Applicability): For this to survive weeks, months, or years in a real university setting, the architecture must integrate with existing infrastructures. The next logical evolution of this system must include Canvas API sync for seamless gradebook integration and Neo4j for complex, cross-course prerequisite mapping.